Reliable CMOS PUF Cell for IoT Hardware Security

Digital Integrated Circuits Project Handout

Course	Digital Integrated Circuits
Project title	Reliable CMOS PUF Cell for IoT Hardware Security
Main tool	Electric VLSI for schematic and layout
Group size	Minimum 3 students; maximum 4 students per project
Submission deadline	Last day of classes: PPT summary and IEEE-style paper. Publishable-track groups may extend to the final exam day after the first paper review.
Issued date	June 2, 2026

Project theme: Design a compact, layout-ready CMOS PUF cell and evaluate whether Schmitt-trigger-assisted readout and unstable-bit screening improve reliability.

Publication ambition: The circuit is intentionally small, but the analysis should be rigorous enough to support a Q1-journal-style manuscript draft.

Student Group Information

Rule: Each project group must include at least 3 students and at most 4 students. Individual projects are not allowed unless approved by the instructor.

#	Student name	Student ID	Role in project
1			
2			
3			
4			

Project Overview

This project challenges students to design a small but research-inspired digital integrated circuit: a compact CMOS physically unclonable function (PUF) cell for low-cost IoT hardware security. A PUF acts as a digital fingerprint for an IC by using natural transistor mismatch. The project therefore connects core digital IC topics with a modern security application: CMOS inverter design, SRAM-like cross-coupling, process variation, noise margin, layout matching, Monte Carlo simulation, and reliability analysis.

Target design idea: A 16-bit Schmitt-trigger-assisted SRAM PUF with lightweight unstable-bit detection.

Reference Paper and Journal Target

Students should read the reference paper carefully and use it as a technical inspiration, not as a circuit to copy exactly.

Main paper: Yan He, Dai Li, Zhanghao Yu, and Kaiyuan Yang, "ASCH-PUF: A 'Zero' Bit Error Rate CMOS Physically Unclonable Function With Dual-Mode Low-Cost Stabilization," IEEE Journal of Solid-State Circuits, 2023.

Useful links: Rice SIMS Lab paper page; arXiv summary; IEEE JSSC submission page; IEEE template selector

The Q1 journal target is IEEE Journal of Solid-State Circuits (JSSC). A more student-friendly Q1 alternative is IEEE Open Journal of the Solid-State Circuits Society (OJ-SSCS). Students should prepare the written report using the IEEE Transactions two-column journal template.

Learning Objectives

- Explain how transistor mismatch can be used to generate a chip-unique response.
- Design and simulate CMOS transistor-level circuits in Electric and SPICE.
- Create a compact full-custom layout and discuss area, matching, and routing tradeoffs.
- Compare a baseline 6T SRAM PUF against an improved Schmitt-trigger-assisted version.
- Present results in a research-paper style with evidence, figures, waveforms, and tables.

Design Challenge

Build a small publishable-inspired version of a reliable SRAM PUF. The circuit should be simple enough for a class project, but the analysis should be strong enough to resemble early-stage journal work.

- **Baseline circuit:** conventional 6T SRAM PUF cell or small SRAM PUF array.
- **Proposed circuit:** 6T SRAM PUF with Schmitt-trigger-assisted sensing or readout.
- **Reliability method:** unstable-bit detection using repeated power-up or repeated-read comparison.
- **Array size:** minimum 4 x 4 PUF array for excellent work; at least one complete PUF bit cell for minimum passing work.
- **Optional extension:** simple XOR or post-processing stage to improve response balance.

Required Circuit Blocks

1. CMOS inverter with correct PMOS/NMOS sizing and voltage-transfer behavior.
2. 6T SRAM cell used as the baseline PUF cell.
3. Schmitt trigger or Schmitt-trigger-inspired read/sensing cell.
4. 1-bit PUF readout circuit.
5. 4 x 4 PUF array with simple row and column selection.
6. Output latch or register.
7. Unstable-bit detector using repeated read or repeated power-up comparison.

Electric Implementation Requirements

- Create clean hierarchical schematics in Electric for each reusable block.
- Use meaningful cell names, signal names, power labels, and ground labels.
- Implement transistor-level layouts for the key cells and array.
- Use clean routing, consistent power rails, contacts, wells, and matching-aware placement.
- Run design-rule checks where available and document any remaining issues.
- Export SPICE/netlist files where possible for simulation and comparison.

Simulation and Analysis Requirements

- Show functional waveforms for the baseline cell and the proposed cell.
- Demonstrate PUF bit generation from power-up or readout behavior.
- Compare the baseline and proposed designs using measurable results.
- Perform mismatch analysis, Monte Carlo simulation, or parameter sweeps when the available model permits.
- Report area, power or energy estimate, bit error rate, reliability, and response balance.
- If possible, run temperature and VDD sweeps and discuss stability under changing conditions.

Required Deliverables

Deliverable	Required content	Due
Electric design files	Schematic cells, layout cells, library file, exported netlists if available.	Last day of classes
Simulation package	Waveforms, testbenches, Monte Carlo or sweep setup, and summarized results.	Last day of classes

Deliverable	Required content	Due
IEEE-style paper	Two-column manuscript using the IEEE Transactions journal template. Include figures, tables, references, and a clear contribution claim.	Last day of classes
Presentation	PPT summary, 8 to 12 slides, with circuit diagram, layout screenshots, simulations, and comparison table.	Last day of classes
Publishable-track revision	Improved paper after instructor review for groups with strong results.	May be extended to final exam day

Report Requirements

- Use the IEEE Transactions two-column journal template.
- Include a clear title, author list, abstract, introduction, related work, proposed design, simulation setup, layout, results, comparison, conclusion, and references.
- All figures must be readable: schematic diagrams, layout screenshots, waveforms, and comparison tables.
- Do not claim fabricated silicon measurements unless the chip was fabricated and measured.
- Clearly state whether results are schematic simulation, layout-extracted simulation, or estimated values.

Research Question

Can a very small SRAM PUF improve reliability without heavy error-correction hardware by using Schmitt-trigger-based readout and lightweight unstable-bit detection?

Evaluation Criteria

Category	Points	Excellent performance	Evidence to submit
Paper understanding	10	Explains the reference paper, problem, circuit idea, limitations, and what is reused or improved.	Short literature summary with correct citation.
Project definition and novelty	10	States a clear research question and identifies the baseline and proposed improvement.	Problem statement and contribution paragraph.
CMOS schematic design in Electric	15	Correct transistor-level schematics with clean hierarchy, sizing, labels, and block reuse.	Electric schematics and screenshots.
Functional simulation	15	Shows correct PUF operation, readout behavior, timing, and baseline/proposed comparison.	Waveforms and simulation setup.
Layout implementation in Electric	15	Complete and compact layout with clean routing, rails, contacts, wells, and matching awareness.	Layout screenshots and area report.

Category	Points	Excellent performance	Evidence to submit
Layout verification	10	DRC clean or justified issues; schematic-layout consistency; extracted simulation if possible.	DRC notes, extracted netlist, verification summary.
Variation and reliability analysis	10	Monte Carlo, mismatch, VDD, temperature, or noise analysis with bit-stability discussion.	Plots, tables, and reliability metrics.
Results comparison	7	Compares conventional 6T SRAM PUF vs proposed design using area, power, reliability, and BER.	Comparison table and discussion.
Report quality	5	IEEE-style writing with readable figures, tables, references, and clear technical claims.	Final two-column paper.
Presentation and defense	3	Student group can defend choices, limitations, and next steps.	Final presentation and Q&A.

Minimum, Excellent, and Bonus Expectations

Level	Expected evidence
Minimum passing work	One working PUF bit-cell schematic, functional simulation with stable digital output, at least one cell or small layout in Electric, and a short report explaining results.
Excellent work	4 x 4 or larger PUF array, baseline vs proposed comparison, mismatch or Monte Carlo simulation, layout area calculation, unstable-bit detection, and an IEEE-style manuscript with publishable figures.
Bonus work: up to 10 points	Post-layout extracted simulation, temperature and VDD sweep, randomness tests, compact area optimization, or a clear claim that could become a journal contribution.

Suggested Project Timeline

Stage	Work to complete	Checkpoint
1	Read the paper, define baseline and proposed circuit, assign group roles.	Short proposal approved.
2	Build and simulate inverter, 6T SRAM cell, and Schmitt trigger.	Waveforms reviewed.
3	Build 1-bit PUF readout and unstable-bit detector.	Functional result shown.
4	Create 4 x 4 array layout in Electric and run verification.	Layout reviewed.
5	Run comparison, reliability analysis, and prepare IEEE-style report and PPT.	Final submission.

Important Rules

- Every group member must understand the complete circuit, not only one isolated block.
- The report must distinguish clearly between copied reference ideas, student design choices, and new proposed improvements.
- All external diagrams, text, and results must be cited. Direct copying from papers, websites, or AI tools is not acceptable.
- The highest grades will go to projects that compare, measure, and argue using evidence: baseline circuit, proposed circuit, simulations, layout, and a clear improvement claim.